

# CVS-RFFI: Cross-Receiver Generalization and Few-Shot Incremental Registration under a Spaceborne Deployment Proxy

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**Abstract**—Radio-frequency fingerprint identification (RFFI) is evaluated with a fixed receiver, transmitter labels, and a closed class set. A spaceborne targetlink monitor violates these assumptions: its receiver is absent from ground training, received IQ undergoes LEO propagation, and unseen transmitters must be enrolled from few labeled records under onboard memory and update constraints. We present CVS-RFFI, a protocol-governed framework that treats ground generalization and deployment registration as equal stages. Phase 1 learns asymmetric 160-dimensional identity and receiver-nuisance representations from proxy receivers under a limited-label split. Receiver-day-gated pseudo-labeling, tail-aware angular geometry, and identity-preserving challenges train the encoder before it and INT8 aggregates are sealed. Phase 2 admits only this immutable bundle and  $K$  labeled target-support records per class. Its RTB-IDR module compiles a 288-dimensional multimodal support representation into one equal-prior affine classifier using perturbation-aware robust centers, task-balanced shrinkage covariance, support-cross-fitted geometry fusion, and residual quantization. Each physical record passes once through a fixed simulated LEO channel; target queries never update the state. A historical Phase 1 audit on a WiSig/ManySig terrestrial proxy reports 89.18% overall accuracy, 84.89% strict unseen-domain accuracy, and a 75.55% receiver floor. In a 125-row matched Phase 2 diagnostic, task-balanced covariance improves post-registration old-class accuracy by 2.62 percentage points and the old/new harmonic mean by 0.96 points for  $K = 10$  with 20 new classes, but reduces new-class accuracy by 0.65 points and provides no gain at  $K = 1$ . The study is a simulation-based prototype, not in-orbit or flight-software validation.

**Index Terms**—radio-frequency fingerprint identification, specific emitter identification, cross-receiver generalization, domain adaptation, few-shot class-incremental learning, satellite communications, physical-layer identity assurance.

## I. Introduction

**R**ADIO-FREQUENCY fingerprint identification (RFFI), also known as specific emitter identification, infers transmitter identity from hardware-dependent distortions introduced by oscillators, mixers, power amplifiers, and other nonideal radio-frequency components. Early radiometric systems used engineered transients and statistics [1]; later work showed that convolutional models can learn discriminative fingerprints directly from complex baseband samples [2]–[4]. Because a hardware fingerprint is tied to the analog emission chain rather than a network identifier, it can provide an independent physical-layer identity cue.

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Commercial broadband LEO constellations give this problem an operational scale. The FCC’s 2022 partial grant authorized an initial tranche of up to 7,500 Gen2 Starlink satellites at 525–535 km, with user-terminal and gateway earth-to-space links included in the system authorization [5]. SpaceX describes its Direct to Cell payloads as orbital cellular base stations that connect ordinary LTE phones and IoT devices, while explicitly identifying Doppler, latency, low antenna gain, and low terrestrial transmit power as link challenges [6]. Measurements of operational Starlink service report short satellite dwell, handovers, and substantially greater time variation than terrestrial access [7]; independent RF studies have recovered the structure of its 10.7–12.7 GHz Ku-band downlink [8]. These facts make high-density LEO networks a useful systems exemplar, not a dataset claim: the experiments in this paper use no Starlink waveform, terminal, satellite, or proprietary interface.

Our sensing direction is the uplink. A receiver aboard a satellite observes terrestrial terminals, gateways, or other authorized emitters and must associate each received waveform with a physical transmitter. In that role, RFFI can supply an auxiliary provenance signal: it can corroborate a claimed terminal identity, help associate interference or replay with a physical emitter, and support enrollment when only a few labeled contacts are available. It does not replace cryptographic authentication. SatiQ shows that hardware fingerprints can raise the cost of spoofing and replay against satellite downlinks [9]; Tang et al. use a 16-USRP proxy to study lightweight onboard identification of TT&C ground stations [10]; a recent survey places satellite-based emitter fingerprinting alongside geolocation for access control, interference mitigation, and wide-area source identification [11]. These studies establish the security role and sensing direction, but not the cross-receiver incremental lifecycle studied here.

The observable waveform is not a transmitter-only object. Propagation, receiver filtering, local oscillators, automatic gain control, sampling clocks, and preprocessing jointly transform the same emitted signal. A model can therefore achieve high same-receiver accuracy by encoding a receiver-channel shortcut rather than a stable transmitter characteristic [12], [13]. This failure is exposed when the receiver changes: cross-receiver RFFI is consistently more difficult than same-receiver classification [14]–[16].

A spaceborne monitoring lifecycle compounds receiver

shift with label, uplink, class, and resource constraints. Ground training cannot observe the final receiver chain, and transmitter labels are costly even in the source domains. At deployment, satellite geometry, residual synchronization errors, multipath, and noise alter the baseband process; elevation, path length, and Doppler are central non-terrestrial-link variables [17]. Short contact windows limit labeled observations, while satellite computing remains constrained by moving infrastructure, power, compute, storage, and the operating environment [18]. These conditions favor ground training followed by compact inference state and sparse onboard updates [19]. The class set is also open to enrollment: a newly encountered transmitter must be added from a small support set without erasing previously registered identities.

Three research lines address these difficulties separately. Cross-receiver RFFI uses collaborative receivers, adversarial disentanglement, distribution alignment, or pseudo-labeling [15], [16], [20]–[22], but normally retains a fixed transmitter set and may use target records for alignment or fine-tuning. Few-shot class-incremental SEI adds devices with prototypes, subnet expansion, self-training, or regularization [23]–[27], but its base and incremental sessions are not designed around a simultaneously unseen receiver and changed ground-to-satellite propagation process. Satellite fingerprinting establishes feasibility on real downlinks [9], [28], while the closest onboard uplink study uses one receiver and a fixed 16-transmitter class set [10]. Ground-satellite generative learning addresses sample scarcity [19]. None of these evaluations jointly tests source-label scarcity, receiver transfer, old-class adaptation, and new-class registration under a sealed deployment state.

The unresolved gap is therefore the intersection of the constraints, not any one constraint in isolation. Among the studies surveyed in Section II, we found no common evaluation that simultaneously enforces: an unseen target receiver; transmitter-label scarcity during ground learning;  $K$ -shot old- and new-class support after deployment; no runtime source IQ, clean counterpart, or target-query feedback; a compact immutable deployment state; and one role-blind predictor over all registered classes. Relaxing any one of these conditions changes the task. Target-batch alignment no longer measures source-only generalization, source replay no longer measures self-contained deployment, and separate old/new heads no longer measure a deployable all-class decision.

We ask whether an RFFI system can first learn an unseen-receiver-tolerant identity space when at most 10% of the source training pool is transmitter-labeled and then adapt old identities and register new ones from  $K$  fixed received records per class, without replaying source samples or retraining on queries. This question requires two coequal stages. Phase 1 must build a representation whose transmitter geometry survives receiver and channel variation under label-limited supervision. Phase 2 must turn scarce support on the deployed receiver into a balanced old/new decision rule within a bounded state.

CVS-RFFI implements this lifecycle. Phase 1 uses an asymmetric dual-representation CV-SincNet: one branch integrates temporal, spectral, and power-amplifier evidence for transmitter identity, while a second branch absorbs receiver, channel, and noise nuisance. Receiver-day-gated semi-supervision, tail-risk geometry, and identity-preserving source challenges train the two representations before the encoder and quantized multi-sample aggregates are sealed. Phase 2 uses RTB-IDR (Robust Task-Balanced Incremental Discriminant Registration) to estimate a receiver-sensitive perturbation spectrum, robustify support centers, balance old and new tasks during covariance estimation, and compile all registered classes into one compact affine head. The channel simulator is a reproducible residual baseband operator, not evidence of an operational satellite link.

The contributions are fourfold.

- 1) We formalize the combined research gap as a two-stage access contract spanning label-limited source-only receiver generalization, a fixed simulated LEO ground-to-satellite observation, support-only old-class adaptation, and few-shot new-class registration under independent all-class inference.
- 2) For Phase 1, we develop an asymmetric identity- nuisance CV-SincNet that couples physical signal views with explicit receiver-information allocation, receiver-day-gated pseudo-labeling, tail-aware angular risk, and an identity-preserving source-domain curriculum. The output is a sealed encoder plus auditable INT8 aggregate knowledge rather than replayable source data.
- 3) For Phase 2, we develop RTB-IDR, a support-only compiler that combines learned and deterministic views, perturbation-aware robust centers, task-balanced shrinkage geometry, bounded Fisher residuals, and residual quantization in one equal-prior affine classifier.
- 4) We report evidence at its supported strength. A historical 32-candidate Phase 1 audit identifies a source-only checkpoint pending confirmation under the current split, while a complete 125-row Phase 2 diagnostic isolates a task-balancing effect that reduces old-class forgetting at large registration scales but does not solve one-shot adaptation or cross-role calibration.

The remainder of the paper reviews related work, formalizes the protocol, describes the two-stage method, reports the evidence currently available, and lists the experiments required before submission-grade performance claims can be made.

## II. Related Work

### A. Deep RFFI and Channel Robustness

Deep RFFI has progressed from raw-IQ convolutional classifiers [2]–[4] to models that use multisampling [29], spectrograms [30], or learned channel compensation [12]. ORACLE explicitly optimizes transmitter classification

with convolutional networks [4], whereas the channel-dissection study of Al-Shawabka et al. shows that an apparently accurate classifier can learn propagation artifacts rather than stable hardware characteristics [13]. WiSig provides a large-scale test bed with 174 transmitters, 41 receivers, and multiple capture campaigns [14]. We use it only as a terrestrial proxy; the present experiments are not measurements from an on-orbit receiver.

## B. Cross-Receiver Generalization and Adaptation

Receiver mismatch has prompted several complementary approaches. Collaborative receiver-agnostic RFFI exploits multiple receivers [15]. RIEI explicitly separates emitter- and receiver-related information and also studies a federated variant [16]. Other methods use target-domain alignment and pseudo-labeling [20], dynamic distribution alignment [21], or joint channel-robust and receiver-independent learning [22]. These studies build on broader DG and domain-adversarial learning principles [31]–[33].

Our distinction is primarily one of lifecycle and evidence. Phase 1 permits only source receivers. Phase 2 permits labeled target support but excludes target-query feedback and runtime source samples. Consequently, a method that uses target data during model selection is classified as adaptation, not DG; a comparison method that requires source replay remains a valid external comparator but is reported under a different permission row.

## C. RFFI over Satellite Links: Downlink Evidence and Uplink Deployment

Satellite RFFI has two sensing directions. PAST-AI and SatIQ identify satellite transmitters from downlinks captured by terrestrial receivers [9], [28]. Tang et al. study the reverse direction: a lightweight CNN-GRU intended for an onboard receiver classifies a 16-USRP proxy for TT&C ground stations [10]. Their experiment uses one receiver, fixed identities, and simulated noise; it neither measures cross-receiver transfer nor enrolls new ground transmitters. A ground-satellite generative framework further studies few-shot satellite IoE identification [19], while the broader satellite sensing literature distinguishes emitter identity from geolocation [11].

Starlink anchors the operational context rather than the evaluation. Its licensed system scale and earth-to-space links [5], Direct to Cell receive direction [6], measured temporal dynamics [7], and observable Ku-band downlink structure [8] show why identity evidence must coexist with moving geometry, handovers, and proprietary waveforms. No Starlink IQ appears in the present corpus. In CVS-RFFI, WiSig/ManySig supplies terrestrial device and receiver observations, and a simulated LEO residual uplink creates clear-sky, low-elevation, and rainy-link baseband profiles. We therefore use “spaceborne deployment proxy” rather than “Starlink validation” or “in-orbit validation” throughout.

## D. Few-Shot and Class-Incremental SEI

Few-shot metric learning commonly represents each class by a support prototype [34]. Class-incremental learning also addresses catastrophic forgetting, often through exemplars, distillation, or classifier expansion [35], [36]. In wireless identification, CSIL expands a device classifier while preserving old knowledge [23]; later methods add self-training and prototype augmentation [24], systematic few-shot incremental design [25], orthogonal or reserved classifier directions [26], and prototype correction with hierarchical regularization [27]. Open-world variants further combine incremental enrollment with rejection or semi-supervised learning [37]–[39].

These methods do not all solve the same problem. Some assume complete base-session training, retain an old teacher, or update a neural network over many gradient steps. CVS-RFFI instead studies a sealed pre-trained backbone followed by support-only registration on a new receiver. Its current RTB-IDR realization uses a parametric discriminant state rather than exemplar replay or query-graph inference.

## III. Problem Formulation and Protocol

### A. Compound Identification Task

Let  $y \in \mathcal{Y}$  denote a transmitter identity,  $d \in \mathcal{D}$  a receiver domain, and  $\chi \in \mathcal{C}_{\text{prop}}$  a propagation condition. We write a length- $N$  complex baseband observation as

$$\mathbf{x}_{y,d,\chi} = \mathcal{R}_d[\mathcal{H}_\chi(\mathcal{T}_y(\mathbf{s}))] + \mathbf{n}, \quad (1)$$

where  $\mathcal{T}_y$  represents transmitter-dependent hardware distortion,  $\mathcal{H}_\chi$  propagation and synchronization effects,  $\mathcal{R}_d$  the receiver chain, and  $\mathbf{n}$  additive noise. The objective is to preserve information about  $y$  while suppressing shortcuts tied to  $d$  and incidental  $\chi$ . Equation (1) is a task decomposition rather than an identifiable inverse model: CVS-RFFI neither reconstructs clean IQ nor estimates a physical power-amplifier model at deployment. Memory-polynomial analysis [40] motivates stable nonlinear transmitter effects, but the classifier operates on the received record.

The deployment problem combines domain shift and class-set expansion. In the intended data flow, terrestrial devices are the transmitters, the target receiver and RFFI inference service are onboard the satellite, authorized labeled contacts form support, and later uplink observations are independently classified as queries. Source training observes proxy receivers  $\mathcal{R}_s$  and old identities  $\mathcal{Y}_o$ . Deployment observes a disjoint receiver set  $\mathcal{R}_t$ , retains the old classes, and registers  $\mathcal{Y}_n$ , where

$$\mathcal{R}_s \cap \mathcal{R}_t = \emptyset, \quad \mathcal{Y}_o \cap \mathcal{Y}_n = \emptyset. \quad (2)$$

The final predictor is a single map over  $\mathcal{Y}_o \cup \mathcal{Y}_n$ ; it is not given an old/new query role. This distinguishes the task from closed-set DG alone, target adaptation alone, and conventional class-incremental learning under source replay.

## B. Simulated LEO Ground-to-Satellite Channel

Equation (1) describes the intended physical uplink, but the proxy corpus does not expose  $\mathcal{T}_y(\mathbf{s})$  or an orbital receiver chain separately. The simulator therefore starts from an already captured terrestrial record  $\mathbf{x}_i^{\text{cap}}$  and applies a post-capture residual overlay:

$$\tilde{\mathbf{x}}_i = \mathcal{A}_{\chi_i}(\mathbf{x}_i^{\text{cap}}; \boldsymbol{\xi}), \quad \chi_i \in \mathcal{C}_{\text{LEO}}, \quad (3)$$

where  $\mathcal{A}_{\chi_i}$  approximates residual ground-to-satellite base-band effects after bulk synchronization and modeled AGC. It neither removes the terrestrial propagation and receiver effects already present in  $\mathbf{x}_i^{\text{cap}}$  nor synthesizes a clean satellite waveform. The three manuscript-facing profiles are

$$\mathcal{C}_{\text{LEO}} = \{\text{CS, LE, RL}\}, \quad (4)$$

denoting clear-sky, low-elevation, and rainy-link conditions, respectively. Given  $\mathbf{x}_i^{\text{cap}} = [x_i[0], \dots, x_i[N-1]]$ , the implemented operator is

$$u_i[n] = \sum_{\ell=0}^1 h_{i,\ell} x_i[n - \delta_{i,\ell}], \quad (5)$$

$$q_i[n] = u_i[n] \exp \left\{ j \left( \frac{2\pi \Delta f_i n}{f_s} + \phi_i[n] \right) \right\}, \quad (6)$$

$$\tilde{x}_i[n] = \frac{a_i q_i[n]}{\sqrt{\max \left\{ N^{-1} \sum_{m=0}^{N-1} |q_i[m]|^2, \epsilon_{\text{ch}} \right\}}} + w_i[n]. \quad (7)$$

Here,  $\delta_{i,0} = 0$ , samples with negative indices are zero, the second delay is sampled from a bounded integer interval,  $a_i = 10^{g_i/20}$  is a residual AGC gain, and  $\epsilon_{\text{ch}} = 10^{-12}$ . The noise obeys  $w_i[n] \sim \mathcal{CN}(0, \sigma_{w,i}^2)$ , with  $\sigma_{w,i}^2 = P_i 10^{-\gamma_i/10}$ , where  $P_i$  is the post-AGC signal power and  $\gamma_i$  the sampled SNR in decibels. The phase process is Wiener:

$$\phi_i[n] = \phi_i[n-1] + \varepsilon_i[n], \quad \varepsilon_i[n] \sim \mathcal{N}(0, \sigma_{\phi,i}^2), \quad (8)$$

and the residual carrier offset follows

$$\Delta f_i \sim \mathcal{N}(0, \sigma_{f,\chi_i}^2). \quad (9)$$

Bulk orbital Doppler is assumed to have been removed by synchronization; the simulator injects only  $\Delta f_i$ .

The dominant tap is Rician or shadowed Rician [41]:

$$h_{i,0} = \sqrt{\rho_0} \left[ \eta_i \sqrt{\frac{\kappa_i}{\kappa_i + 1}} e^{j\psi_i} + \sqrt{\frac{1}{\kappa_i + 1}} z_{i,0} \right], \quad (10)$$

$$h_{i,1} = \sqrt{\rho_1} z_{i,1}, \quad (11)$$

where  $z_{i,0}, z_{i,1} \sim \mathcal{CN}(0, 1)$ ,  $\psi_i \sim \mathcal{U}(-\pi, \pi)$ , and  $(\rho_0, \rho_1) = (1, p_{\chi_i})/(1+p_{\chi_i})$ . For the low-elevation profile,  $\eta_i = 10^{v_i/20}$  with  $v_i \sim \mathcal{N}(0, 1.5^2)$  dB; otherwise  $\eta_i = 1$ . The Rician factor varies linearly with elevation:

$$\kappa_i^{\text{dB}} = \kappa_{\min, \chi_i} + (\kappa_{\max, \chi_i} - \kappa_{\min, \chi_i}) \frac{\theta_i - \theta_{\min, \chi_i}}{\theta_{\max, \chi_i} - \theta_{\min, \chi_i}}. \quad (12)$$

The gain equations use  $\kappa_i = 10^{\kappa_i^{\text{dB}}/10}$ .

Following standard LEO geometry [17], altitude  $h_i$  and elevation  $\theta_i$  determine the slant range

$$r_i = -R_E \sin \theta_i + \sqrt{(R_E + h_i)^2 - R_E^2 \cos^2 \theta_i} \quad (13)$$

TABLE I  
Common Simulated LEO Channel Settings

| Parameter           | Setting                          | IQ role      |
|---------------------|----------------------------------|--------------|
| Record length $N$   | 256 complex samples              | Applied      |
| Sampling rate $f_s$ | 25 MS/s                          | Applied      |
| Carrier $f_c$       | 2.462 GHz                        | Metadata     |
| Altitude $h$        | $\mathcal{U}(500, 2000)$ km      | Metadata     |
| Multipath           | Two taps; $\delta_0 = 0$         | Applied      |
| Bulk Doppler        | Compensated before overlay       | Not injected |
| Absolute FSPL       | Retained in record metadata      | Not applied  |
| Atmospheric loss    | Disabled                         | Not applied  |
| Extra IQ imbalance  | Disabled                         | Not applied  |
| AGC target RMS      | 1, then residual $g$             | Applied      |
| Realization policy  | Role-seeded stream; sealed bytes | Applied      |

and free-space path loss

$$L_{\text{FS},i} = 20 \log_{10} \left( \frac{4\pi r_i f_c}{c_0} \right). \quad (14)$$

Here,  $R_E = 6.371 \times 10^6$  m and  $c_0 = 299\,792\,458$  m/s. The present residual-channel implementation records  $r_i$  and  $L_{\text{FS},i}$  as metadata but does not multiply absolute path loss into IQ, because every record is RMS-normalized by the modeled AGC. Carrier frequency and altitude therefore affect recorded geometry metadata, not the output IQ. Likewise, explicit atmospheric attenuation and additional IQ imbalance are disabled. The rainy-link profile represents the post-AGC SNR, fading, phase-noise, and residual-frequency regime in Table II; it is not a full link-budget rain-attenuation model.

## C. Phase 1: Source-Only Learning

Phase 1 uses the labeled source set

$$\mathcal{L}_s = \{(\mathbf{x}_i, y_i, d_i) : d_i \in \mathcal{R}_s\} \quad (15)$$

and the unlabeled source set

$$\mathcal{U}_s = \{(\mathbf{x}_j, d_j) : d_j \in \mathcal{R}_s, y_j \text{ hidden}\}. \quad (16)$$

The protocol defines the transmitter-labeled fraction within the source training pool as

$$\rho_{\text{label}} = \frac{|\mathcal{L}_s|}{|\mathcal{L}_s| + |\mathcal{U}_s|} \leq 0.10. \quad (17)$$

Its current governed split allocates 0.07, 0.63, and 0.30 of the full source pool to labeled training, unlabeled training, and disjoint source validation, so  $0.07/(0.07+0.63) = 0.10$ . No record from  $\mathcal{R}_t$  may affect training, normalization statistics, early stopping, thresholds, prototypes, or checkpoint selection. Phase 1 outputs an immutable bundle  $\mathcal{B}_{\text{P1}}$  containing the selected encoder and licensed aggregate state, but no target-derived quantity. The historical checkpoint audited in Section V-B used a 0.10/0.70/0.20 full-pool split: 10% of the full source pool, but 12.5% of its training pool, carried transmitter labels. It is therefore internal method evidence rather than the final protocol-locked confirmation.

TABLE II  
Scenario-Specific Parameters of the Simulated Residual LEO Channel

| Parameter                          | Clear-sky (CS)               | Low-elevation (LE)                          | Rainy-link (RL)                             | Sampling rule / unit               |
|------------------------------------|------------------------------|---------------------------------------------|---------------------------------------------|------------------------------------|
| Elevation $\theta$                 | 35–90                        | 10–35                                       | 20–80                                       | Uniform, degrees                   |
| SNR                                | 22–32                        | 16–28                                       | 14–26                                       | Uniform, dB                        |
| Residual CFO $\sigma_f$            | 50                           | 90                                          | 70                                          | Gaussian std., Hz                  |
| Phase-increment std. $\sigma_\phi$ | $0\text{--}5 \times 10^{-4}$ | $1 \times 10^{-4}\text{--}8 \times 10^{-4}$ | $1 \times 10^{-4}\text{--}7 \times 10^{-4}$ | Uniform, rad/sample                |
| Fading law                         | Rician                       | Shadowed Rician                             | Rician                                      | LOS shadow std. in LE: 1.5 dB      |
| $\kappa^{\text{dB}}$ range         | 16–24                        | 8–18                                        | 10–20                                       | Linear in elevation                |
| Second-tap delay $\delta_1$        | 1–2                          | 1–3                                         | 1–3                                         | Discrete uniform, samples          |
| Second-tap factor $p_\chi$         | 0.08                         | 0.12                                        | 0.10                                        | Powers normalized as $[1, p_\chi]$ |
| Residual AGC $g$                   | −0.2–0.2                     | −0.3–0.3                                    | −0.3–0.3                                    | Uniform, dB                        |

#### D. Phase 2: Fixed Observation and Support-Only State

Before Phase 2, each physical IQ record produces exactly one allowed observation through Eq. (3). The current capsule builder uses a role-seeded ordered random-number stream: the two-tap delay pattern is shared within a processed batch, while the remaining draws advance in builder order. Consequently, a record is not independently regenerated from its physical identifier and the recorded role seed alone. The authoritative replay surface is the validated capsule containing the frozen scenario assignment and output bytes. Equalization, normalization, FFTs, and other mathematical views may read  $\tilde{\mathbf{x}}_i$ , but they do not create additional physical samples. Thus,  $K$ -shot means  $K$  distinct physical records, not  $K$  transformations of one record.

Let the old and newly enrolled class sets be  $\mathcal{Y}_o$  and  $\mathcal{Y}_n$ , with  $\mathcal{Y}_o \cap \mathcal{Y}_n = \emptyset$ . Every registered class  $c \in \mathcal{Y} = \mathcal{Y}_o \cup \mathcal{Y}_n$  has a labeled support set

$$\mathcal{S}_c = \{(\tilde{\mathbf{x}}_{c,k}, c)\}_{k=1}^K. \quad (18)$$

Support and query physical identifiers are disjoint. A deployment state is compiled as

$$\Theta = \text{Compile}\left(\mathcal{B}_{\text{P1}}, \bigcup_{c \in \mathcal{Y}} \mathcal{S}_c, \Gamma\right), \quad (19)$$

where  $\mathcal{B}_{\text{P1}}$  is an immutable Phase 1 bundle and  $\Gamma$  is a method configuration frozen before query evaluation.

For every query, the predictor produces one independent decision

$$\hat{y}_j = \arg \max_{c \in \mathcal{Y}} s_c(\tilde{\mathbf{x}}_j^{(q)}; \Theta). \quad (20)$$

The predictor cannot read the query label, its true old/new role, the number of examples from each class in the query batch, or another query's features. It cannot enforce class quotas or use Hungarian assignment, optimal transport, label propagation over queries, or global reassignment. Labels are joined only after immutable predictions are emitted.

TABLE III  
Phase 2 Access and Claim Semantics

| Stage | Target information                                                   | Permitted claim                                   |
|-------|----------------------------------------------------------------------|---------------------------------------------------|
| 2-A   | No target transmitter labels; optionally unlabeled fixed received IQ | Zero-label target reference only                  |
| 2-B   | $K$ -shot labeled support for $\mathcal{Y}_o$                        | Old-class target adaptation and calibration       |
| 2-C   | $K$ -shot labeled support for $\mathcal{Y}_o \cup \mathcal{Y}_n$     | Joint old-class retention and seen-new enrollment |

#### E. Metrics

Let  $A_o^{\text{pre}}$  be old-class accuracy after Stage 2-B and before adding new classes,  $A_o^{\text{post}}$  old-class accuracy after Stage 2-C, and  $A_n$  seen-new accuracy. We report

$$F = A_o^{\text{pre}} - A_o^{\text{post}} \quad (21)$$

and

$$H_{\text{old,new}} = \frac{2A_o^{\text{post}}A_n}{A_o^{\text{post}} + A_n}. \quad (22)$$

We also report the lowest old-class accuracy within each row, receiver-wise results, and scenario-wise results. A valid Stage 2-C claim must keep  $A_o^{\text{post}}$ ,  $A_n$ , and  $H_{\text{old,new}}$  from the same experimental row.

### IV. Proposed CVS-RFFI Framework

#### A. Lifecycle Overview

CVS-RFFI does not treat Phase 1 as generic pre-training or Phase 2 as an auxiliary classifier. Phase 1 determines which transmitter evidence remains stable when the receiver changes; Phase 2 determines how that evidence can be recalibrated and extended when only target support is available. The target query is absent from both state-construction processes. Figure 1 shows the resulting ground-to-deployment boundary.

#### B. Phase 1: Label-Limited Cross-Receiver Representation Learning

1) Asymmetric Physical Multi-View Encoder: The frozen Phase 1 model receives  $\mathbf{x} \in \mathbb{R}^{2 \times 256}$ . A shared front

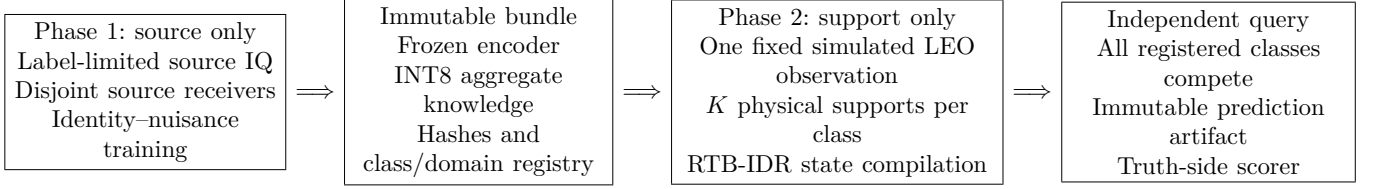


Fig. 1. The CVS-RFFI lifecycle. The target query is absent from training and registration. The independent scorer receives ground truth only after predictions are sealed.

end applies 24 learnable Sinc bandpass filters of length 79 [42] and first- and second-order high-frequency differences. Above this stem, two parameter-distinct backbones produce

$$\mathbf{z}_{\text{id}} = f_{\theta}^{\text{id}}(\mathbf{x}) \in \mathbb{R}^{160}, \quad \mathbf{z}_{\text{dom}} = f_{\theta}^{\text{dom}}(\mathbf{x}) \in \mathbb{R}^{160}. \quad (23)$$

The identity backbone combines temporal, spectral, and power-amplifier (PA) evidence. Its PA path explicitly lifts the received IQ through a short memory polynomial [40],

$$\phi_{r,o}(\mathbf{x})[n] = x[n-r]|x[n-r]|^{o-1}, \quad (24)$$

$$r \in \{0, 1, 2, 3\}, \quad o \in \{1, 3, 5\},$$

before gated convolution and projection. Temporal, spectral, circularity, and PA features are fused into the final identity vector, which is the only learned embedding used for Phase 2 class geometry.

The domain backbone is deliberately asymmetric. It disables style mixing, retains a widely linear complex path for DAC/IQ conjugate coupling, and augments its learned domain feature with 18 receiver-channel-noise statistics. Let  $\mathbf{r}_{\text{rcn}}(\mathbf{x})$  collect these statistics and let  $\mathbf{z}_{\text{dom}}^0$  denote the learned domain feature. The implemented gated enhancement is

$$\mathbf{e}_{\text{rcn}} = E_{\text{rcn}}(\mathbf{r}_{\text{rcn}}(\mathbf{x})), \quad (25)$$

$$\mathbf{g}_{\text{rcn}} = \sigma(\mathbf{W}_g[\mathbf{z}_{\text{dom}}^0; \mathbf{e}_{\text{rcn}}] + \mathbf{b}_g), \quad (26)$$

$$\mathbf{z}_{\text{dom}} = \text{LN}(\mathbf{z}_{\text{dom}}^0 + 0.35 \mathbf{g}_{\text{rcn}} \odot \mathbf{e}_{\text{rcn}}). \quad (27)$$

Conversely, the identity branch removes the DAC path and applies same-transmitter cross-domain MixStyle [32] only to labeled temporal features. Training executes both branches; deployment bypasses the domain branch. This construction allocates nuisance evidence to a separate path, but the shared low-level stem means that statistical independence is neither assumed nor claimed.

2) Receiver-Nuisance Allocation: The identity head uses CosFace [43]. For transmitter class  $c$ , its frozen logit is

$$\ell_{i,c}^{\text{tx}} = 30 \left[ \frac{(\mathbf{z}_i^{\text{id}})^{\text{T}} \mathbf{w}_c}{\|\mathbf{z}_i^{\text{id}}\|_2 \|\mathbf{w}_c\|_2} - 0.35 \mathbf{1}[c = y_i] \right], \quad (28)$$

and the supervised transmitter loss uses 0.01 label smoothing. Receiver/day domain labels then play two opposite roles. A domain head predicts  $d_i$  from  $\mathbf{z}_{\text{dom}}$ , whereas

a gradient-reversal head [31] predicts  $d_i$  from  $\mathbf{z}_{\text{id}}$  while reversing the gradient entering the identity backbone:

$$\mathcal{L}_{\text{dom}} = -\frac{1}{B} \sum_i \log p_{\phi}(d_i | \mathbf{z}_i^{\text{dom}}), \quad (29)$$

$$\mathcal{L}_{\text{adv}} = -\frac{1}{B} \sum_i \log p_{\psi}(d_i | \text{GRL}(\mathbf{z}_i^{\text{id}})). \quad (30)$$

Two additional terms prevent the two heads from satisfying their labels through unrelated shortcuts. For centered batch matrices  $\tilde{\mathbf{Z}}_{\text{id}}$  and  $\tilde{\mathbf{Z}}_{\text{dom}}$ ,

$$\mathcal{L}_{\text{orth}} = \frac{1}{160^2} \left\| \frac{\tilde{\mathbf{Z}}_{\text{id}}^{\text{T}} \tilde{\mathbf{Z}}_{\text{dom}}}{B-1} \right\|_{\text{F}}^2, \quad (31)$$

$$\mathcal{L}_{\text{cons}} = \frac{1}{|\mathcal{P}|} \sum_{(c,d,d') \in \mathcal{P}} [1 - \cos(\boldsymbol{\mu}_{c,d}, \boldsymbol{\mu}_{c,d'})],$$

where  $\boldsymbol{\mu}_{c,d}$  is the normalized identity center of transmitter  $c$  in source domain  $d$ , and  $\mathcal{P}$  contains available same-transmitter, different-domain center pairs. The scheduled separation loss is

$$\mathcal{L}_{\text{sep}}(e) = w_{\text{dom}}(e) \mathcal{L}_{\text{dom}} + w_{\text{adv}}(e) \mathcal{L}_{\text{adv}} + w_{\text{orth}}(e) \mathcal{L}_{\text{orth}} + w_{\text{cons}}(e) \mathcal{L}_{\text{cons}}. \quad (32)$$

Its maximum weights are 1, 0.35, 0.05, and 0.08, respectively. Domain observability, adversarial suppression, cross-covariance reduction, and class-conditional alignment are thus optimized together; none alone proves complete disentanglement.

3) Receiver-Day-Gated Semi-Supervision and Tail Risk: In the audited checkpoint, 10% of the full source pool carries transmitter labels, while 70% is available without transmitter labels and 20% is reserved for source validation; equivalently, 12.5% of the historical source training pool is labeled under Eq. (17). During epochs 131–200, an exponential-moving-average (EMA) teacher with decay 0.999 predicts a lightly perturbed unlabeled record. For confidence  $\kappa_j = \max_c q_{j,c}$ , the receiver-day threshold is

$$\tau_d = \text{clip}(Q_{0.86}\{\kappa_j : d_j = d\}, 0.92, 0.97). \quad (33)$$

A pseudo-label is admitted only if it clears this threshold, the teacher's domain prediction agrees with the known source domain, the adjacent time-window decision is consistent, and the student agrees under a stronger per-

turbation. With the three agreement indicators denoted by  $g_j^{\text{dom}}, g_j^{\text{temp}}, g_j^{\text{str}}$ ,

$$m_j = \mathbf{1}[\kappa_j \geq \tau_{d_j}] g_j^{\text{dom}} g_j^{\text{temp}} g_j^{\text{str}}, \quad (34)$$

$$\mathcal{L}_u = - \frac{\sum_j m_j \log p_\theta(\hat{y}_j | a_s(\mathbf{u}_j))}{\sum_j m_j + \epsilon}. \quad (35)$$

An entropy term is evaluated on all strong-view predictions. Receiver-day conditioning prevents an easy domain from setting the acceptance pattern for every receiver; the gates control label noise rather than increasing the nominal number of labeled records.

Mean source accuracy can still hide a collapsed receiver or transmitter. For each source domain  $g$ , the model computes its label-smoothed transmitter loss  $\ell_g$  and averages only the largest 35%:

$$\mathcal{L}_{\text{group}} = \frac{1}{K_g} \sum_{g \in \text{Top}_{K_g}(\{\ell_h\})} \ell_g, \quad K_g = \lceil 0.35|\mathcal{G}| \rceil. \quad (36)$$

A domain-wise logit-gradient-variance penalty aligns optimization variability across source domains. The identity space is further constrained by cross-epoch prototypes, cross-domain supervised contrast, a 40° angular radius, class-balanced tail risk, core–tail preservation, and soft interclass boundary mixtures. These are closed-set source-domain objectives; they do not constitute unknown-class training.

4) Identity-Preserving Source Challenges: Phase 1 creates source-only challenges for which the transmitter label must remain invariant. Same-transmitter cross-domain MixStyle starts with probability 0.18, shape parameter 0.10, and strength 0.70; after epoch 110, probability and strength anneal over 40 epochs to 0.05 and 0.32. A leave-one-source-domain episode constructs an explicit held-domain classification problem within each labeled batch. From epoch 80, labeled records also receive a simulated residual LEO channel view and retain their original transmitter label:

$$\mathcal{L}_{\text{LEO}} = - \frac{1}{B} \sum_i \log p_\theta(y_i | a_{\text{LEO}}(\mathbf{x}_i)). \quad (37)$$

The active coefficient is 0.68. The frozen checkpoint assigns zero weight to clean-to-LEO consistency; only the classification loss in (37) contributes a gradient. MixStyle, episodic DG, and LEO augmentation are established mechanisms individually. Their role here is the identity-preserving coordination of receiver style, held-source extrapolation, and deployment-channel stress.

5) Scheduled Objective and Immutable Bundle: For readability, the implemented nonzero terms are grouped as

$$\begin{aligned} \mathcal{L}_{\text{P1}}(e) = & \mathcal{L}_{\text{tx}} + \mathcal{L}_{\text{sep}}(e) + \mathcal{L}_{\text{risk}}(e) + \lambda_{\text{epi}}(e) \mathcal{L}_{\text{epi}} \\ & + 0.68 \mathbf{1}[e \geq 80] \mathcal{L}_{\text{LEO}} \\ & + \mathbf{1}[e \geq 131] (0.16 \mathcal{L}_u + 0.01 \mathcal{L}_{\text{ent}}). \end{aligned} \quad (38)$$

Here,  $\mathcal{L}_{\text{risk}}$  groups the nonzero hard-domain, gradient-variance, prototype, angular, tail, and soft-boundary

TABLE IV  
Phase 1 Optimization Schedule

| Epochs  | Data stage          | Dominant addition                                                                 |
|---------|---------------------|-----------------------------------------------------------------------------------|
| 1–16    | Labeled             | Core TX/domain learning; partial adversarial, orthogonal, and hard-domain weights |
| 17–68   | Labeled             | Ramp receiver suppression, class-conditional alignment, and tail geometry         |
| 69–79   | Labeled             | Full separation and source-domain risk weights                                    |
| 80–130  | Labeled             | Add simulated LEO transmitter classification                                      |
| 131–200 | Labeled + unlabeled | Add gated pseudo-label CE and entropy minimization                                |

terms described above; their individual coefficients and warm-up epochs are frozen before target access. Table IV summarizes the dominant schedule. Style mixing acts inside the identity feature path and is therefore not written as an additive loss.

Checkpoint selection uses only disjoint source validation and a frozen joint criterion over overall, strict unseen-domain, and receiver-floor behavior. The resulting deployment bundle contains the checkpoint, architecture and normalization metadata, class/domain registries, and 84 quantized domain–class aggregate identity centers. Each center is formed from at least two distinct Phase 1 physical records before target access and stored with an INT8 code, scale, and validity mask. The bundle contains no raw source IQ, single-sample feature, reversible index, or full-precision exemplar. Phase 2 reads but cannot update it. Changing this aggregate state changes the bundle identifier, not the already validated target-data capsule.

### C. Phase 2: Support-Only Incremental Registration

Phase 2 addresses the complementary deployment problem with the Phase 1 encoder and bundle frozen. RTB-IDR receives labeled target support for all currently registered old and new classes, performs no gradient update, and compiles a single state used by every subsequent query. The procedure cannot access source records, clean counterparts, query features from other decisions, or query truth.

1) Multimodal Target Feature: For each fixed received record, RTB-IDR computes three views from the same physical observation:

$$\mathbf{a} = \mathcal{N}_\epsilon(\mathbf{z}_{\text{id}}) \in \mathbb{R}^{160}, \quad (39)$$

$$\mathbf{b} = \mathcal{N}_\epsilon([\mathbf{f}_{\text{FFT96}}; \mathbf{f}_{\text{RF32}}]) \in \mathbb{R}^{128}, \quad (40)$$

where  $\mathcal{N}_\epsilon(\mathbf{v}) = \mathbf{v} / \max(\|\mathbf{v}\|_2, \epsilon)$  and  $\epsilon = 10^{-8}$ . FFT96 is obtained by centering, applying a Hann window, taking the log-magnitude spectrum, and interpolating to 96 bins. RF32 contains fixed amplitude, phase, moment, imbalance, and short-lag correlation statistics. Neither view increases  $K$ .

The joint feature is

$$\mathbf{z} = \mathcal{N}_\epsilon \left( \begin{bmatrix} \mathbf{a} \\ 4\mathbf{b} \end{bmatrix} \right) \in \mathbb{R}^{288}. \quad (41)$$

For nonzero blocks, their final norm contributions are  $1/\sqrt{17}$  and  $4/\sqrt{17}$ . The fixed factor 4 is part of the locked candidate and is not selected from target-query accuracy.

2) Perturbation-Aware Robust Centers: The INT8 source aggregates are dequantized, centered within each class across source domains, and pooled into a perturbation covariance  $\mathbf{G}$ . After symmetric correction and subtraction of the estimated quantization-noise floor,

$$\mathbf{G}_+ = \frac{\mathbf{G} + \mathbf{G}^\top}{2} - \sigma_q^2 \mathbf{I}. \quad (42)$$

The positive eigenspectrum  $(\lambda_j, \mathbf{u}_j)$  defines a class-agnostic basis  $\mathbf{U}$ . Its rank is the ceiling of the positive-spectrum participation ratio, so rank selection does not use target-query results.

For class  $c$ , let  $\bar{\mathbf{z}}_c^{\text{id}}$  be the mean identity support and  $\mathbf{e}_{c,k}$  the residual of support  $k$ . The perturbation energy is

$$E_{c,k} = \sum_{j=1}^r \rho_j (\mathbf{u}_j^\top \mathbf{e}_{c,k})^2, \quad \rho_j = \frac{\lambda_j}{\sum_{\ell=1}^r \lambda_\ell}. \quad (43)$$

With  $\tau_c = K^{-1} \sum_k E_{c,k}$ , one-step Cauchy weights are

$$\omega_{c,k} \propto \frac{1}{1 + E_{c,k}/\tau_c}. \quad (44)$$

The weighted identity center defines a translation applied uniformly to all identity supports of class  $c$ ; their within-class residuals and the FFT/RF blocks are unchanged. For  $K \leq 2$  or a degenerate energy scale, the module takes a conservative identity fallback. This prevents the method from fabricating covariance evidence at one shot.

3) Task-Balanced Shrinkage Geometry: The 288-dimensional support covariance is ill-conditioned when  $K$  is small. RTB-IDR therefore uses automatic shrinkage [44] separately within the old and new tasks:

$$\hat{\Sigma}_o = \text{ShrinkCov}(\{\mathbf{z}_{c,k} : c \in \mathcal{Y}_o\}), \quad (45)$$

$$\hat{\Sigma}_n = \text{ShrinkCov}(\{\mathbf{z}_{c,k} : c \in \mathcal{Y}_n\}). \quad (46)$$

After registration, the balanced covariance is

$$\hat{\Sigma}_{\text{bal}} = \frac{1}{2} \hat{\Sigma}_o + \frac{1}{2} \hat{\Sigma}_n. \quad (47)$$

The 0.5/0.5 weights are fixed and do not vary with the number of new classes, receiver, scenario, seed, or transmitter identifier. Before new-class registration, and in the  $K \leq 2$  fallback, the registered-old control geometry is retained exactly.

Two covariance structures are constructed. A full model retains cross-block correlations among identity, FFT, and RF features; a block model zeros the cross-block terms. Leave-one-shot-rank cross-fitting estimates class-wise reliability without query access. The resulting fusion is compiled into a single score matrix; it is not a query-time mixture chosen from a query's old/new role.

4) Bounded Fisher Residual and Affine Compilation: A low-rank Fisher residual [45] is estimated from support class means and within-class scatter. Candidate residuals are clipped and admitted only through support-cross-fitted per-class and atomic safety checks. Failed candidates leave the base geometry unchanged. Equal class priors are then used to compile a linear discriminant head

$$\mathbf{w}_c = \hat{\Sigma}^{-1} \boldsymbol{\mu}_c, \quad (48)$$

$$b_c = -\frac{1}{2} \boldsymbol{\mu}_c^\top \hat{\Sigma}^{-1} \boldsymbol{\mu}_c, \quad (49)$$

$$s_c(\mathbf{q}) = \mathbf{q}^\top \mathbf{w}_c + b_c. \quad (50)$$

All old and new classes use the same score form and equal prior. Thus, task balancing affects covariance estimation but does not create separate old/new heads.

5) Quantized State: Each affine coefficient block is stored with two residual INT8 codes and two FP16 scales; intercepts use FP16 and the diagonal metric uses FP32. For  $C$  registered classes and  $p = 288$ ,

$$B_{\text{core}} = 4p + 2Cp + 14C \text{ bytes}. \quad (51)$$

The 26-class core arrays occupy 16.11 KiB, and the compiled head requires  $Cp = 7488$  multiply-accumulate operations per query. These figures exclude the frozen encoder, FFT/RF extraction, registration workspace, metadata, and the current FP32 decode path.

## V. Experimental Methodology

### A. Data and Splits

We use the WiSig/ManySig corpus as a terrestrial receiver/transmitter proxy. The public WiSig dataset contains 10 million packets from 174 transmitters and 41 receivers across four captures [14]. The frozen CVS-RFFI subset uses seven source receivers

$$\{1-1, 1-19, 14-7, 18-2, 19-2, 2-1, 2-19\}$$

and five disjoint target receivers

$$\{20-1, 3-19, 7-14, 7-7, 8-8\}.$$

Six transmitter identities appear in Phase 1 and are old classes in Phase 2. Nested sets of 5, 10, and 20 additional identities are used as new classes. Every target record is assigned one of {CS, LE, RL} before support/query selection. The received record length, sampling rate, carrier frequency, geometry, fading, synchronization residuals, and noise ranges are fixed by Tables I and II.

[AUTHOR ACTION: Insert the exact public data-release identifier, local manifest hash, and per-split physical-record counts from the frozen capsule.]

### B. Phase 1 Selection Audit

The Phase 1 audit contains 32 completed candidates. Each has a complete 200–240 epoch trace, checkpoint, diagnostic export, and source-only evaluation. The selected checkpoint, ADV3B02\_CORE90\_SOFT\_E200, uses 200 epochs and is selected at epoch 194. It is compared

with the preceding ADV2 aggregate rather than with an untraceable cross-run maximum. This historical audit used the 0.10/0.70/0.20 full-source-pool split implemented by that run, corresponding to a 12.5% labeled fraction inside its training pool. A final confirmation must use the current 0.07/0.63/0.30 protocol split, whose training-pool labeled fraction is 10%; Table V is not relabeled as evidence from that unexecuted split.

We report overall source-domain accuracy, strict unseen-domain/unseen-split (UDU) accuracy, the minimum receiver-level accuracy (receiver floor), and the strict floor under satellite-stress augmentation. The stress metrics remain proxy metrics and do not certify target adaptation.

### C. Phase 2 Matched Matrix

The RTB-IDR diagnostic fixes five target receivers, seeds 713102–713106, and five slices:

$$(K, C_n) \in \{(1, 20), (5, 20), (10, 5), (10, 10), (10, 20)\}.$$

Each receiver–seed–slice row evaluates all three simulated LEO profiles, yielding 125 jobs and 375 scenario cells. The authoritative retry completed 125/125 jobs with zero execution failures.

The matched control, internally named D81, shares the frozen features, robust centers, dual-geometry fusion, safety gates, support/query sets, and seeds. The only controlled difference is whether Eq. (47) is activated after new-class registration. The paired deltas therefore isolate task-balanced covariance; they do not estimate the contribution of the entire RTB-IDR pipeline relative to a raw backbone.

### D. External Comparisons

We include official-semantics reproductions of CSIL [23] and MoPC-HR [27]. Their complete matrix contains 5 receivers, 5 seeds, 4 values of  $K$ , 4 new-class scales, and two methods, for 800 cells and 2400 scenario rows. The new-class support and query records use the same simulated LEO channel convention, but the methods retain the base/source access required by their original training lifecycles. Their results are therefore descriptive, not paired with RTB-IDR.

[AUTHOR ACTION: Before submission, add same-capsule, same-seed, same- $C_n$  support-only ProtoNet, single-qKNN, adapter-qKNN, and a frozen current CVS candidate. Report external-paper methods in a separate permission table rather than merging incompatible rows into one leaderboard.]

### E. Evidence and Statistical Policy

All adaptations are constructed from support only. Hyperparameters are locked before query scoring. Each query prediction is written to an immutable artifact, and a separate scorer joins truth afterward. Main interpretations use complete same-row tuples; marginal maxima from

TABLE V  
Phase 1 Source-Only Results (%)

| Metric                 | ADV2 avg. | CVS-RFFI P1 | Difference |
|------------------------|-----------|-------------|------------|
| Overall                | 86.92     | 89.18       | +2.26      |
| Strict UDU             | 80.09     | 84.89       | +4.80      |
| Receiver floor         | 69.07     | 75.55       | +6.48      |
| Satellite strict floor | 67.83     | 68.77       | +0.94      |

different receivers or seeds are not combined. Means are descriptive unless a paired interval is available.

[AUTHOR ACTION: Compute receiver-clustered or receiver-by-seed bootstrap confidence intervals and paired effect intervals from the final frozen confirmation matrix. The current Phase 1 comparison has no independent-repeat interval and must not be described as statistically significant.]

## VI. Results

### A. Phase 1 Source-Only Generalization

Table V shows that the selected Phase 1 checkpoint improves the mean, strict split, and worst-receiver criteria simultaneously. The receiver-floor gain is the largest, suggesting that model selection based only on overall accuracy would understate the benefit. The satellite-stress floor improves by less than one point, however, and remains below a separate soft-unknown baseline’s 69.55%. The selected checkpoint also has a source-overflow diagnostic of 0.459 and a bridge-accept diagnostic of 1.0. These diagnostics prevent an open-set claim: the Phase 1 evidence supports closed-set source-only DG, not unknown rejection.

### B. Matched Effect of Task-Balanced Covariance

Table VI yields three conclusions. First, the task-balanced covariance is inactive at  $K = 1$ , as required by the identifiability fallback; its paired effect is exactly zero. Second, as the new-class set grows, it increasingly protects old classes. At  $K = 10, C_n = 20$ , all five target receivers improve in old-class accuracy, and the gains under clear, low-elevation, and rain stress are 2.00, 2.77, and 3.10 points. Third, the gain is not Pareto-complete. The same large-scale row loses 0.65 points of new-class accuracy, and the  $K = 10, C_n = 5$  row slightly favors new classes at the expense of old-class accuracy and floor.

The absolute values also matter. At  $K = 10, C_n = 20$ ,  $H_{\text{old,new}} = 69.56\%$ , and the minimum old-class accuracy is 42.67%. Thus, complete execution and a positive paired delta do not imply deployment success.

### C. Descriptive External Comparison

RTB-IDR obtains a substantially higher harmonic mean than the two reproduced incremental methods in Table VII. This is not a clean algorithm ranking. The old-pre values reveal that the methods enter registration with very different base representations, and their source/base

TABLE VI  
RTB-IDR Absolute Results and Paired Change Versus the Matched Control

| $K$ | $C_n$ | $A_o^{\text{pre}}$ | $A_o^{\text{post}}$ | Min-old | $A_n$  | $H_{\text{old,new}}$ | $F$    | $\Delta A_o^{\text{post}}$ | $\Delta \text{Min-old}$ | $\Delta A_n$ | $\Delta H_{\text{old,new}}$ |
|-----|-------|--------------------|---------------------|---------|--------|----------------------|--------|----------------------------|-------------------------|--------------|-----------------------------|
| 1   | 20    | 68.144             | 44.033              | 14.200  | 27.150 | 33.410               | 24.111 | 0.000                      | 0.000                   | 0.000        | 0.000                       |
| 5   | 20    | 81.267             | 63.711              | 33.200  | 58.883 | 60.955               | 17.556 | +2.311                     | +2.400                  | -0.410       | +0.920                      |
| 10  | 5     | 86.111             | 76.189              | 49.800  | 74.133 | 74.803               | 9.922  | -0.133                     | -0.867                  | +0.520       | +0.197                      |
| 10  | 10    | 86.111             | 72.533              | 44.200  | 66.353 | 69.106               | 13.578 | +1.000                     | +1.933                  | -0.340       | +0.291                      |
| 10  | 20    | 86.111             | 71.333              | 42.667  | 68.150 | 69.555               | 14.778 | +2.622                     | +4.600                  | -0.653       | +0.964                      |

TABLE VII  
Different-Permission Comparison at  $K = 10, C_n = 20$  (%)

| Method               | Old-pre | Old-post | New    | $H$    |
|----------------------|---------|----------|--------|--------|
| RTB-IDR              | 86.111  | 71.333   | 68.150 | 69.555 |
| CSIL reproduction    | 42.833  | 38.222   | 1.660  | 2.979  |
| MoPC-HR reproduction | 45.322  | 7.611    | 25.187 | 10.695 |

TABLE VIII  
RTB-IDR Compiled-State Resource Profile

| $C$ | FP32 affine | Core state | Head MAC/query |
|-----|-------------|------------|----------------|
| 6   | 6.77 KiB    | 4.58 KiB   | 1,728          |
| 11  | 12.42 KiB   | 7.46 KiB   | 3,168          |
| 16  | 18.06 KiB   | 10.34 KiB  | 4,608          |
| 26  | 29.35 KiB   | 16.11 KiB  | 7,488          |

permissions, training schedules, and seeds differ. The result instead shows that original incremental-learning recipes can become severely imbalanced when transferred to simulated LEO cross-receiver data with many simultaneously enrolled classes.

We intentionally exclude an older two-new-class qKNN table from this ranking. Changing  $C_n$  changes the candidate set and the cross-role confusion problem; comparing its  $H$  with a 5-, 10-, or 20-new-class row would be invalid.

#### D. Resource Profile

The compiled state is small, but registration is not free. The current  $K = 10$  audit performs approximately 88 component fits and has an 11.15–11.74 GMAC-equivalent upper-bound accounting, together with FP64 covariance workspaces. The Python/NumPy/scikit-learn reference implementation decodes INT8 weights to FP32 before the final product. Accordingly, Table VIII supports a storage claim for the head, not an end-to-end INT8 acceleration claim.

### VII. Discussion

#### A. What Is Novel and What Is Not

The individual ingredients of CVS-RFFI have precedents: Sinc filters [42], cosine-margin classification [43], domain-adversarial learning [31], style perturbation [32], shrinkage covariance [44], and Fisher discrimination [45]. The defensible novelty lies in their task-specific lifecycle and coupling:

- 1) an explicit separation between source-only semi-supervised receiver DG and support-only incremental deployment;
- 2) an immutable, quantized aggregate source-knowledge boundary rather than runtime source replay;
- 3) a class-permutation-symmetric state compiler that uses identical formulas for old and new supports, yet prevents new-class count from dominating covariance estimation; and
- 4) an evidence path in which every query is independently scored over all registered classes and truth is attached only after prediction publication.

The present evidence establishes these properties and a targeted forgetting reduction. It does not yet establish a state-of-the-art end-to-end accuracy result. For a top-tier Transactions submission, the protocol contribution must be accompanied by a stronger final Stage 2 method or by a broader benchmark release that makes the protocol itself independently reusable.

#### B. Operational Role and Security Boundary

In a deployed uplink receiver, CVS-RFFI would be one sensor in a larger identity and spectrum-assurance chain. Its defensible functions are to corroborate a credentialed transmitter, maintain physical-device provenance across contacts, prioritize anomalous emissions for review, and compile a small registered-class state after authorized support arrives. A closed-set RFFI score alone cannot prove malicious intent, distinguish an unknown emitter, or justify autonomous access denial. Those actions require calibrated rejection, attack-specific validation, and integration with cryptographic and network-layer evidence.

Starlink motivates the scale, mobility, and contact dynamics of the problem; it is not the evaluated system. The current model has not processed a Starlink uplink, executed on a Starlink-compatible payload, or been tested against a Starlink terminal. Any system-specific claim would require compatible RF captures, receiver-front-end characterization, a complete link budget, attack and benign controls, and target-processor measurements. The present evidence supports a protocol and a terrestrial-to-LEO proxy, not a security assessment of Starlink.

#### C. Failure Analysis

The  $K = 1$  result exposes a structural limit: one support per class cannot identify within-class perturbation or

estimate task-specific covariance. Query-assisted pseudo-labeling could add apparent samples, but it would violate the present support-only contract and would confound adaptation with evaluation.

At larger  $K$ , Eq. (47) balances covariance evidence, not score scales. New classes can still intrude into old-class decision regions. A licensed role-oracle diagnostic at  $K = 10$ ,  $C_n = 20$  increases old accuracy from 71.33% to 83.31% and  $H$  from approximately 69.7% to 76.9%. The oracle is permanently non-promotable, but the gap localizes the remaining problem: legal cross-role calibration must be learned from support, not from a query's true role.

#### D. Validity Boundaries

Proxy-to-space gap: WiSig/ManySig is terrestrial. The simulated LEO operators are residual baseband proxies, not orbital measurements or a complete link-budget model. Claims about radiation, uncompensated Doppler dynamics, satellite payload electronics, rain attenuation, or flight reliability require real hardware and real links.

Selection multiplicity: Phase 1 was selected from 32 candidates. The reported comparison is an internal audit and lacks a fresh, independently repeated confirmation. Its deltas should not be interpreted as unbiased generalization gains until reproduced under a frozen selection rule.

External-method permissions: CSIL and MoPC-HR are reproduced according to their own lifecycles. Their low CVS results do not invalidate their original papers. A fair manuscript must show both original-task fidelity and the permission mismatch when transferring them to CVS.

Incremental-lifecycle evidence: The current matrix performs one joint registration event over the available old- and new-class supports. It does not test a sequence of independently arriving enrollment sessions, persistent-state updates, arrival-order sensitivity, or rollback. The evidence therefore supports few-shot class-set expansion in one sealed event, not a completed continual onboard lifecycle.

Resource evidence: The final head is compact, but the complete registration pipeline has no worst-case execution time, peak resident memory, energy, thermal, or radiation-tolerance measurement on a target processor.

Table IX separates the current evidence lock from the measurements required for a Q1 submission. In particular, neither compact head storage nor completed diagnostic execution substitutes for end-to-end onboard validation or a frozen promotable Stage 2 result.

### VIII. Conclusion

This paper introduced CVS-RFFI, a protocol-governed framework for cross-receiver RFFI under a space-borne deployment proxy. It separates source-only dual-representation learning from support-only old-class adaptation and new-class enrollment, and it makes the deployment boundary auditable through fixed received observations, immutable model knowledge, independent all-class decisions, and truth-side scoring. The current RTB-IDR

TABLE IX  
Evidence Lock for the Submission Version

| Claim surface        | Current status and required evidence                                                                                                    |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 DG           | Internal audit only; add fresh frozen repeats, intervals, module ablations, and matched cross-receiver baselines.                       |
| Stage 2 registration | Negative diagnostic; add a frozen promotable method on the full $K$ , receiver, seed, scenario, and $C_n$ matrix with same-row metrics. |
| Support baselines    | Existing slices differ; match capsule, seeds, new-class scale, physical IDs, and permission rows for qKNN/ProtoNet.                     |
| Satellite transfer   | Untested; add real-satellite or hardware-in-the-loop validation, or retain proxy-only language.                                         |
| Onboard efficiency   | Head storage only; measure end-to-end latency, peak RAM, energy, and numerical closure on the target processor.                         |
| Reproducibility      | Local evidence only; release code, hashes, split manifest, channel parameters, and a truth-isolated scorer.                             |

realization compiles multimodal target support into a compact discriminant state. Its 125-row matched diagnostic shows that task-balanced covariance can reduce old-class forgetting when many new classes are registered, but the improvement is accompanied by a new-class tradeoff and vanishes at one shot. The evidence therefore supports the protocol, the state-compilation mechanism, and a specific forgetting result; it does not yet support in-orbit validation or a completed state-of-the-art deployment claim. Resolving support-only cross-role calibration, one-shot identifiability, and target-hardware efficiency is the necessary next step.

#### Data and Code Availability

The base WiSig dataset is described in [14]. The current experiment artifacts, protocol capsules, and implementation are maintained in a controlled research repository. [AUTHOR ACTION: Replace this paragraph with the final public repository, immutable release tag, data-access instructions, split-manifest hashes, and artifact archive DOI. Do not promise release terms that have not been approved.]

#### Acknowledgment

[AUTHOR ACTION: Insert funding, compute-resource, data-provider, and contributor acknowledgments after authorship review.]

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